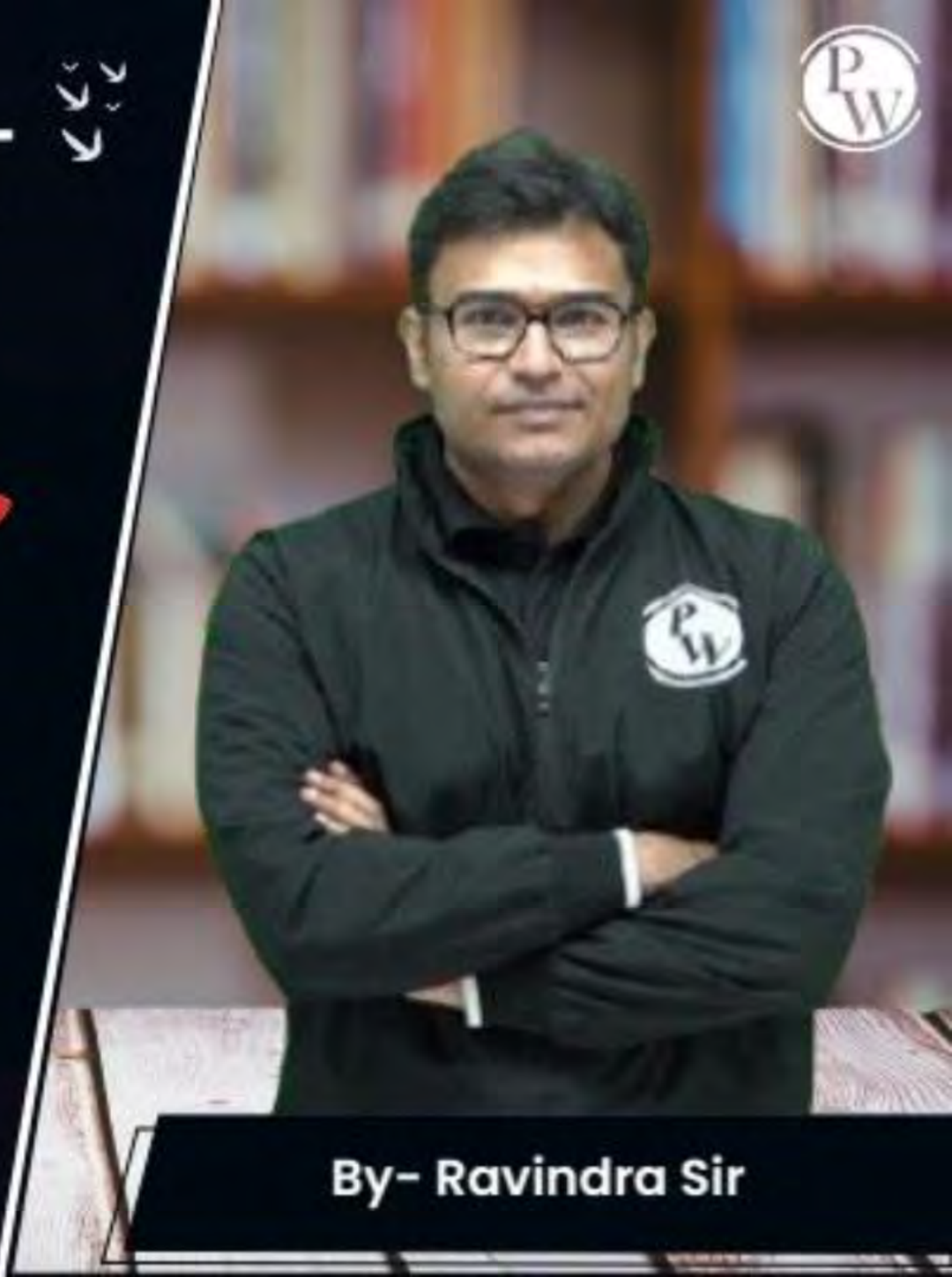
# Computer Science & IT

## Theory of Computation



Regular Languages

Lecture No. 14



By- Ravindra Sir

# Recap of Previous Lecture



- Topic
- Topic
- Topic

closure

# Topics to be Covered



Topic

Topic

Topic

Homo morphism

homomorphism → Substitution

$L = \{01, 10, 11\}$

$h(0) = a$

$h(1) = bb$

$h(L) = \{abb, bba, bbbb\}$

Which of the following languages are not closed under union

a) Reg b) OCFL c) REC d) RE

→ what operations is DCFL closed

a) ∪ b) ∩ c) Comp ✓ d) *

→ $L_1, L_2 \rightarrow$ DCFL

$L_1 \cap L_2 \rightarrow ?$

↑DCFL ∩ ↑DCFL → X

↑CFL ∩ ↑CFL → X

CSL ∩ CSL → (CSL)

$Reg \cup (\underline{DCFL})^R \cap CFL$

Reversal $\rightarrow \cap \rightarrow \cup$

$Reg \cup (CFL)^R \cap CFL$

$CFL \cap CFL$

$CSL \cap CSL$

$CSL \rightarrow CSL$

$Reg \cup$

No language is closed under following operations

- $\subseteq$
- $\supseteq$
- infinite $\cup$
- infinite $\cap$
- infinite set difference

not closed → may or may not be

Reg languages are not closed under Subset operation

$a^n b^n \subseteq a^* b^*$

non Reg ⊆ Reg

$\epsilon \subseteq a^* b^*$

Reg ⊆ Reg

Subset of a reg langage may or may not be Reg

Regular languages are not closed under superset operation

$\phi \subseteq a^n b^n$

$\phi$ → Regular

$a^n b^n$ → non Reg

Superset of a regular language may or may not be regular

$\phi \subseteq a^* b^*$

$\phi$ → Regular

$a^* b^*$ → Regular



$\phi$

## infinite union :-

$L_1 \cup L_2 \cup L_3 \cup L_4 \cdots\cdots$

$\epsilon$ — R

$a$ — R

$aa$ — R

$aaa$ — Re

$\longrightarrow a^* \rightarrow$ Regular ✓

$L_1 \cup L_2 \cup L_3 \cup L_4 \cdots\cdots$

$ab$ — R

$aabb$ — R

$aaabbb$ — R

$aaaabbbb$ — R

$\longrightarrow a^n b^n / n \geq 1 \rightarrow$ non reg

## Secondary operations

$L_1 - L_2 = L_1 \cap \overline{L_2}$

$L_1 \oplus L_2 = (L_1 \cap \overline{L_2}) \cup (\overline{L_1} \cap L_2)$

$L_1 \Rightarrow L_2 = \overline{L_1} \cup L_2$

$L_1 \Leftrightarrow L_2 = (L_1 \cap L_2) \cup (\overline{L_1} \cap \overline{L_2})$

$L_1 \uparrow L_2 = \overline{L_1 \cap L_2}$

$L_1 \downarrow L_2 = \overline{L_1 \cup L_2}$

RE - REC ↑

RE - RE ✗

RE ∩ $\overline{RE}$

Not closed

RE - REC

= RE ∩ REC → REC ↑

= RE ∩ RE

= RE

Note:- First expand the operation, the apply the rules, then push up




✓

DCFL ∪ Reg

CFL ∩ Reg → CFL

CSL - Reg → CSL

Reg – DCFL

Reg ∩ DCFL ✓

Reg ✓ ∩ DCFL ✓

DCFL ✓

CSL ✓ X

DCFL - Reg

(L) - R

DCFL

Reg - CFL

Reg ∩ CFL ✓

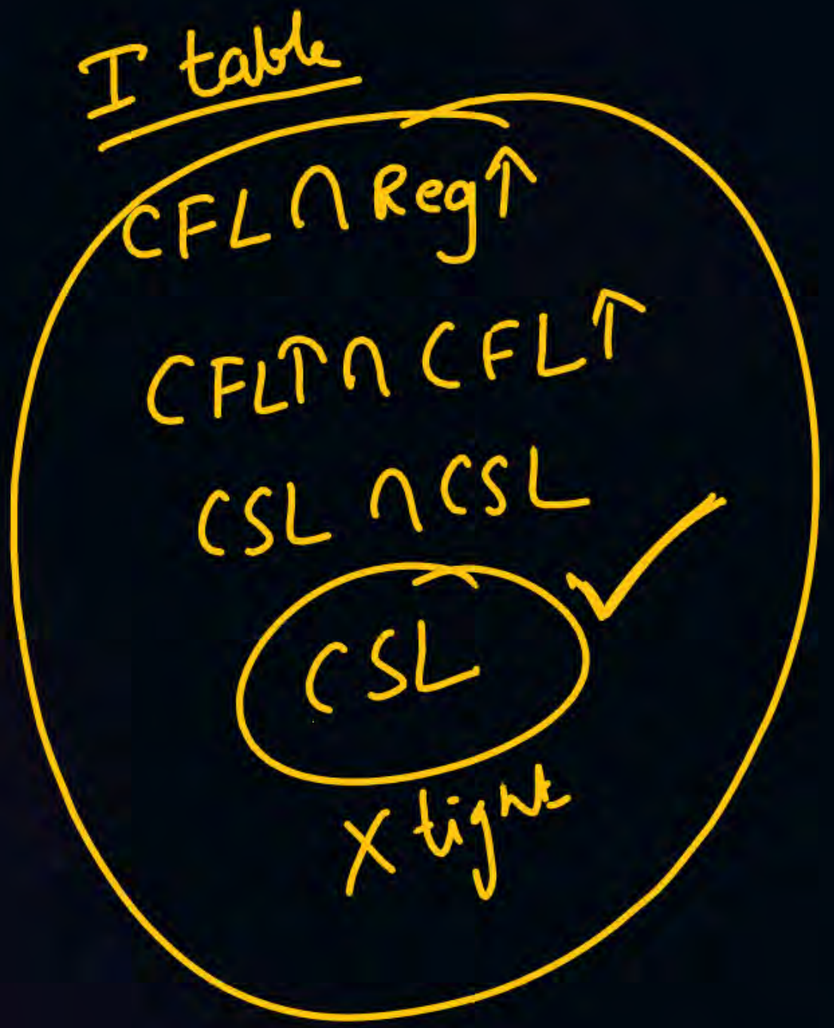
$\overline{CSL}$

Reg ∩ CSL ✓

CSL ✓

Reg ∪, ∩ −


CFL ∩ Reg
CFL


I table
CFL ∩ Reg ↑
CFL↑ ∩ CFL↑
CSL ∩ CSL
CSL
X tight

Reg - Rec

Reg ∩ Rec Rec

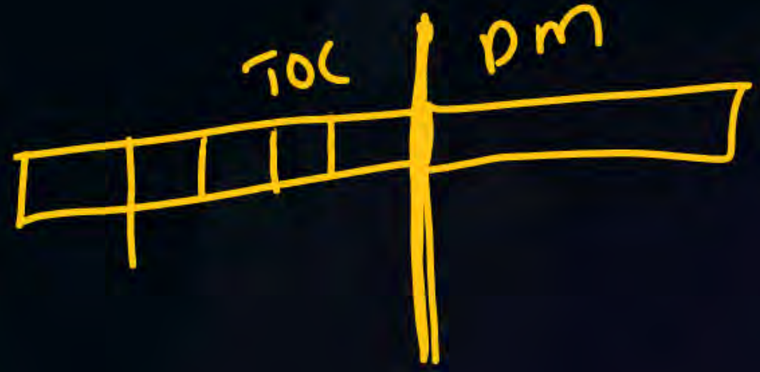
Rec

L-R

R-L

| | Reg | CFL | |
|---|---|---|---|
| ✓ prefix of init(L) | ✓ | ✓ | Suffix |
| ✓ Right quotient ($L_1/L_2$) | ✓ | ✓ | Left quotient |
| Cycle (L) | ✓ | ✓ | |
| min (L) | ✓ | X | |
| max (L) | ✓ | X | |
| Half (L) | ✓ | X | |
| Alt (L) | ✓ | X | |

Not imp ←



## init(L) & prefix(L):-

✓ $L = \{10, 001\}$

$L \subseteq prefix(L)$

10 → ϵ, 1, 10

001 → ϵ, 0, 00, 001

✓ $Prefix(L) = \{\epsilon, 1, 10, 0, 00, 001\}$

$L = a^*$ init(L) & prefix(L)?

$L = \{\epsilon, a, aa, aaa \cdots\}$

$P(L) = \{\epsilon, a, aa, aaa \cdots\} = a^*$

If L is finite, the prefix(L) is finite

If L is infinite, the prefix(L) is infinite.

$L = a^+$ Init(L) = ?

$L = \{a, aa, aaa \cdots\}$

$I(L) = \{\epsilon, a, aa, aaa \cdots\} = a^*$.

$L = (ab)^*$ init(L) = ?

abababab

$\epsilon$, a, ab, aba, abab, ababa, abababab

$(ab)^* + (ab)^*a$

$(ab)^*(\epsilon + a)$

$L = a^*b^*$ $\quad init(L) = ?$

$a^{\circledcirc}b^{\circledcirc} \rightarrow a^*, b^*, a^*b^*, \epsilon$

$L = \{(\epsilon, a, aa, aaa \cdots)$
$(b, bb, bbb \cdots)$
$(ab, aab, abb \cdots)\}$

$I(L) = a^* + b^* \; a^*b^*$

$a^*b^*$

$L = (a+b)^*$ ✓ $I(L) = ?$

$Init(L) \supseteq L$ ✓

$Init(L) = (a+b)^*$

$L = \{a^n b^n / n \geq 1\}$ $I(L) = ?$

$Init(L) \supseteq L$

$Init(L) = a^n b^n / n \geq 1 \cup a^m b^1 / m \geq 1 \cup \{\epsilon\}$

$a^*$

aaabbb

$\epsilon, a, aa, aaa, aaab, aabb, aaabbb$

Note: Init(L) always contains "$\epsilon$" (except $L = \phi$)

$L = \{ w / n_a(w) = n_b(w) \} \neq a^n b^n$

$L = \{ \in, ab, ba, abab, aabb, baba, bbaa \cdots \}$

$I(L) = (a+b)^*$

aabb

ababab

baabba

aba

ababaa

u, b

# Right quotient: $(L_1/L_2)$

$L_1 = \{101, 01\}$

$L_2 = \{01, 1, 0\}$

$L_1/L_2 = \{u / uv \in L_1 \& v \in L_2\}$

$L_1/L_2 = \{11, 110, \epsilon, 0\}$

11 $\frac{1101}{01}$ $\frac{1101}{1}$ $\frac{1101}{110}$ $\frac{1101}{0}$ $\phi$ $\frac{01}{01}$ $\frac{0}{1}$ $\frac{\phi}{01}$ $\frac{\phi}{0}$

Left quotient :- $L_1 \setminus L_2$

$L_1 = \{1101, 01\}$

$L_2 = \{01, 1, 0\}$

$L_1 \setminus L_2$

$\phi$ $\frac{1101}{01}$ $\quad$ 101 $\frac{1101}{\cancel{1}01}$ $\quad$ $\phi$ $\frac{1101}{0}$

$\epsilon$ or $\phi$ $\quad$ $\phi$ $\frac{01}{1}$ $\quad$ 1 $\frac{01}{\phi}$

$(\epsilon, 1, 101)$

$\rightarrow L_1 = 10^* \quad L_2 = \underline{0} \quad L_1/L_2 = ?$

$L_1 = \{\cancel{\phi}, \cancel{1}\cancel{0}, \cancel{10}\cancel{\phi}, \cancel{100}\cancel{\phi}, \cancel{1000}\cancel{\phi} \cdots\cdots \}$

$\phi \qquad 1, 10, 100, 1000 \cdots\cdots$

$10^*$

$\phi \quad \cancel{1}\cancel{0} \quad \cancel{1}\cancel{\phi} \quad \cancel{10}\cancel{\phi} = 10$

$L_1 = a^*$ $L_2 = a$ $L_1 / L_2 = ?$

$$\{\epsilon, \frac{\epsilon}{a}, \frac{a}{a}, \frac{aa}{a}, \frac{aaa}{a}, \frac{aaaa}{a} - \cdots\}$$

$$\epsilon, a, aa, \cdots$$

$$a^*$$

$L_1 = a^+ \quad L_2 = a \quad L_1/L_2 = ?$

$\epsilon \{ \frac{\alpha}{\alpha}, \frac{a\alpha}{\alpha}, \frac{aa\alpha}{\alpha}, \frac{aaa\alpha}{\alpha} \cdots - \}$

$a^*$

$L_1 = 10^*$ $L_2 = 01^*$ $L_1/L_2$

$L_1 = 10^*$

$L_1 = \underline{1}0^*$

$L_2 = \{\underline{0}, \underline{01}, 011, 0111, \cdots\}$

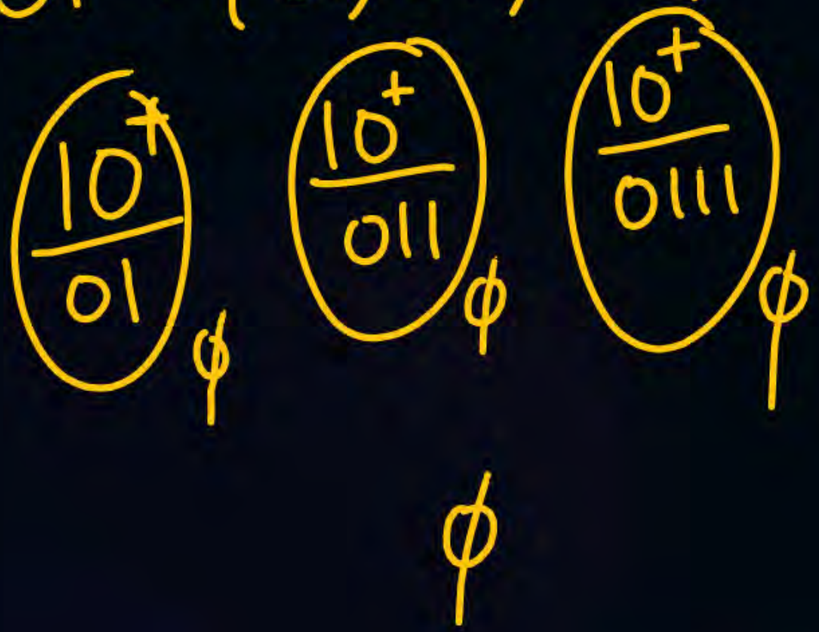
$\frac{10^*}{0}$ $\frac{10^*}{01}\phi$ $\frac{10^*}{011}\phi$ $\frac{10^*}{0111}\phi$

$= 10^*$

$\frac{1}{0}, \frac{10}{0}, \frac{100}{\phi}, \frac{100}{\phi}, \cdots$

$10^+/01^+$

$01^+ = \{01, 011, 0111, \ldots\}$




$\phi$

$\frac{\infty}{\infty}$ = finite & infinite

$\frac{\text{fin}}{\text{fin}}$ = finite

$L_1 = a^n b^n$ $L_2 = (a+b)^*$

$L_1 \cup L_2 = ?$

✓ Regular → Right

Table: $a^n b^n$ ↓ CFL ∪ $(a+b)^*$ ↓ Reg → CFL → Union

→ If $L_1$ is Reg and $L_2$ is Reg then $L_1 \cup L_2$ is Reg

→ If $L_1 \cup L_2$ is Reg then $L_1, L_2$ may or may not be Reg (Proof)

If $L_1 \cap L_2$ is Reg then $L_1, L_2$ may or may not be Reg (Proof)

If $L_1 \cdot L_2$ is Reg then $L_1, L_2$ may or may not be Reg

If $\overline{L_1}$ is Reg, then $L_1$ Reg

If $L_1^R$ is Reg, then $L_1$ is regular

Union :-

| $L_1$ | $L_2$ | |
|---|---|---|
| R | R | |
| $a^*$ | $b^*$ | $L_1 \cup L_2 = a^* + b^*$ (Reg) |
| R | NR | |
| $(a+b)^*$ | $a^n b^n$ | $L_1 \cup L_2 = (a+b)^*$ Reg |
| NR | NR | |
| $a^n b^n$ | $a^n b^n$ | $L_1 \cup L_2 = (a+b)^*$ Reg |

$$L_1 \cup \overline{L_1} = \Sigma^*$$

$$\overline{L_1} = \Sigma^* - L$$

n:

| $L_1$ | $L_2$ | |
|---|---|---|
| R | R | $L_1 \cap L_2 = \{\epsilon\} \rightarrow Reg$ |
| $a^*$ | $b^*$ | |
| R | NR | $L_1 \cap L_2 = \phi \rightarrow Reg$ |
| $\phi$ | $a^n b^n$ | |
| NR | NR | $L_1 \cap L_2 = \phi \rightarrow Reg$ |
| $a^n b^n$ | $a^n b^n$ | |

→ If $L_1$ is regular $\iff$ $\overline{L_1}$ is regular

→ If $L_1$ is not regular $\iff$ $\overline{L_1}$ is not regular

→ $(L_1, \overline{L_1})$ are both regular & both non regular

→ $L_1$ is reg and $L_2$ is regular $\Rightarrow$ $L_1 \cup L_2$ is regular

$$p \Rightarrow q$$

$$\sim q \Rightarrow \sim p$$

If $L_1 \cup L_2$ is non reg $\Rightarrow$ $L_1$ is NR & $L_2$ is NR

| | Reg | DCFL | CSL | REC |
|---|---|---|---|---|
| $\bar{L}$ | ✓ | ✓ | ✓ | ✓ |

L and $\bar{L}$ are both Reg ⇒ L and $\bar{L}$ are both non Reg

L and $\bar{L}$ are both DCFL ⇒ L and $\bar{L}$ are both not= DCFL

L and $\bar{L}$ are both CSL ⇒ L and $\bar{L}$ are both not. CSL

L and $\bar{L}$ are both Rec ⇒ L and $\bar{L}$ are both not Rec

→ If $L^*$ is regular ⇒ L may or may not be regular

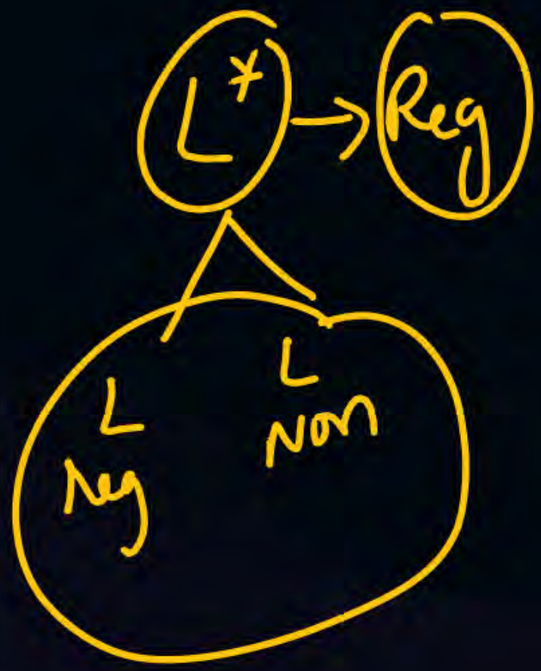
$L = \{a^{n^2} / n \geq 1\}$ → non reg

$L^* = \{@, aaaa, a^8, \ldots\}^*$

$L^* = a^*$

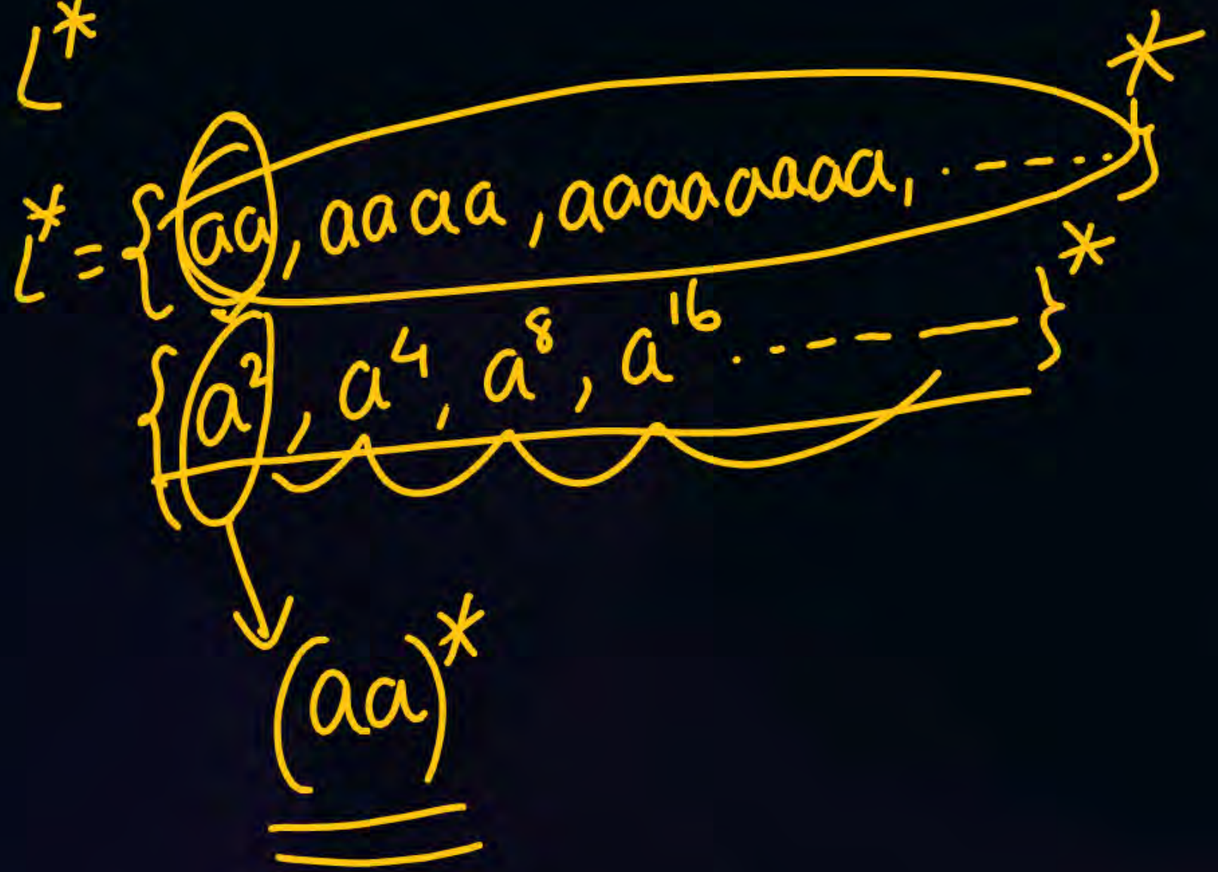
$L = \{a\}$

$L^* = a^*$



$L = \{a^{2^n} / n \geq 1\} \rightarrow$ non reg

$L^*$



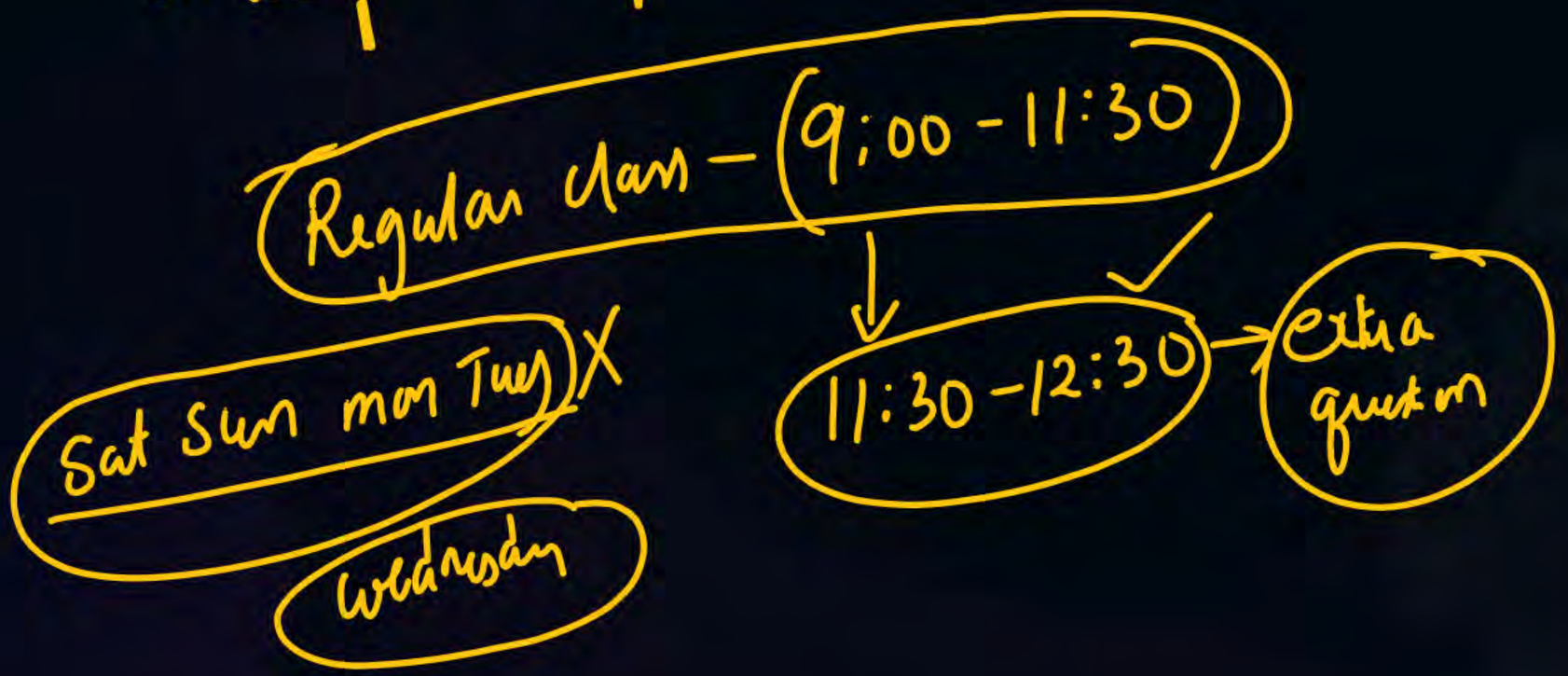
→ If a problem has an algorithm, then it is called decidable

# Decidability table

| | Reg | CFL | CSL | REC | RE |
|---|---|---|---|---|---|
| $(w \in L)$ membership ✓ | D | D | D | D | UD |
| $(L = \phi)$ Emptyness ✓ | D | D | UD | UD | UD |
| ($L$ is finite) finiteness ✓ | D | UD | UD | UD | UD |
| $(L_1 = L_2)$ equivalence ✓ | D | UD | UD | UD | UD |
| $(L = R)$ Regularity ✓ | D | UD | UD | UD | UD |
| ($L$ = ambiguous) ambiguity ✓ | D | UD | UD | UD | UD |
| $(L = \Sigma^*)$ Completeness ✓ | D | UD | UD | UD | UD |
| $(L_1 \cap L_2 = \phi)$ Disjointedness ✓ | D | UD | UD | UD | UD |

Construct a mode machine that takes Set of all strings over {a,b} and prints '1' as o/p for every occurance of 'ab' as a substring

a ab a a ab
  1       1



# THANK - YOU